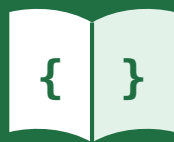

Module 23 — Spring Cloud

Exercises

Spring Boot Edu



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How to Work

1. The starter code is in `modules/23-spring-cloud/exercise/`: a small gateway (Spring Cloud Gateway, WebFlux) in front of a catalog and a review service. Find the TODO comments (also in `application.yaml`).
2. The tests use WireMock for the two services, so they need no Docker:

```
./mvnw -Pexercises -pl modules/23-spring-cloud/exercise -am test
```

3. The exercise is done when all tests are green.
4. If you get stuck, read the hints first, then the solution in `modules/23-spring-cloud/solution/`.

Exercise 1 — A New Route with an API Key (Medium)

Goal: the gateway also serves the review service, and only clients with a valid API key get through.

Tasks:

- TODO 1a (`application.yaml`) — a route `reviews: /api/reviews/**` to the service `review-service` through the load balancer.
- TODO 1b (`security.ApiKeyFilter`) — every request to `/api/**` needs a valid `X-API-Key` header. Otherwise answer 401 and stop.

Hints:

- Lesson section 3.1: `uri: lb://...` and a Path predicate.
- In a `GlobalFilter`: `exchange.getResponse().setStatusCode(HttpStatus.UNAUTHORIZED)` and return `exchange.getResponse().setComplete();` end the request without calling a service.
- The key comes from `bookstore.gateway.api-key (dev-key)`.

Acceptance criteria: all 3 tests in `Exercise1Test` pass.

Estimated time: 30 minutes

Exercise 2 — The Breaker Opens (Medium)

Goal: when the review service fails, the gateway answers with an empty, marked list and, after repeated failures, stops calling it for a while.

Tasks:

- TODO 2a (application.yaml) — a CircuitBreaker filter named reviews on the route of exercise 1, with the fallback forward:/fallback/reviews. 503 and 500 answers count as failures.
- TODO 2b (fallback.FallbackController) — /fallback/reviews answers {"reviews": [], "fallback": true}.

Hints:

- The filter arguments: name, fallbackUri, statusCodes (a list). Without statusCodes, only exceptions and timeouts count, and a 503 from the service goes to the client unchanged.
- The breaker settings (resilience4j.circuitbreaker.instances.reviews) are given: 4 calls, 50 %.
- @RequestMapping("/fallback/reviews") accepts every HTTP method.

Acceptance criteria: both tests in Exercise2Test pass.

Estimated time: 30 minutes

Exercise 3 — Rotate the Key Without a Restart (Easy)

Goal: a leaked API key is replaced in the configuration. The gateway must use the new key after POST /actuator/refresh, without a restart.

Tasks (security.ApiKeys):

- TODO 3 — make the bean read the configuration again after a refresh.

Hints:

- Lesson section 3.5: @RefreshScope.
- The test changes the environment and calls ContextRefresher.refresh(), which is what /actuator/refresh does.

Acceptance criteria: Exercise3Test passes.

Estimated time: 10 minutes

Extra Challenge (Optional)

Put the exercise gateway in front of the compose system of the lesson: add it to compose.yaml on port 9100, with the review route pointing to a small WireMock container (wiremock/wiremock). Then stop the WireMock container and watch the breaker open (/actuator/circuitbreakers).